ODE-FFN: Progress Report 2

Independent Study — Deep Learning

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1 Project and progress

This project investigates whether ODE-inspired transformations provide a useful inductive bias for FFNs in flow-matching models, with JiT as the eventual application. We explored clocks, P, Taylor order, directions and nonlinear responses. Ordinary controls explain several gains. The latest finding is that field fitting and centered residual distributions improve while mode probabilities drift.

Tests isolate the FFN in a width-64, three-block, time-conditioned MLP, learning velocity from $z_t = tX + (1-t)e$ with independent noise–data pairs. Ring8 has eight equal-weight 2D Gaussian modes; Product16 is a rotated product of 16 bimodal factors. Oracle MSE measures fitting; sliced Wasserstein distance (SW2) measures generation. Earlier tests use 5k updates; recent tests use 20k, batch 512 and Heun sampling.

2 Local dynamics: clocks, order and optimization

The original scalar map at $K = 2$ is

$$T(h) = h + s(h)Ph + \tfrac{1}{2}s(h)^2P^2h, \qquad s(h) = 0.2 + 0.8\,\sigma(g(h)/10). \tag{1}$$

Clocks and order. The input clock differs from FM time and contributes to the full Jacobian. On Ring8, input gates reduce samples beyond 0.6 from every mode center: 19.9% to 8.3%, versus SwiGLU's 3.4% (parameter counts differ). Freeing fixed-clock matrices improves fitting. Exact solves, state feedback and clock/augmentation variants do not improve the cross-task quality–cost tradeoff. Scalar K1 improves all eight matched K1/K2 field comparisons. At 80k on Ring8, raw K1 beats matched SwiGLU64 on fitting; SwiGLU wins on sampling and with larger widths.

Optimization. Muon approximately orthogonalizes FFN momentum updates; other parameters retain AdamW. We measured both updates and learned P.

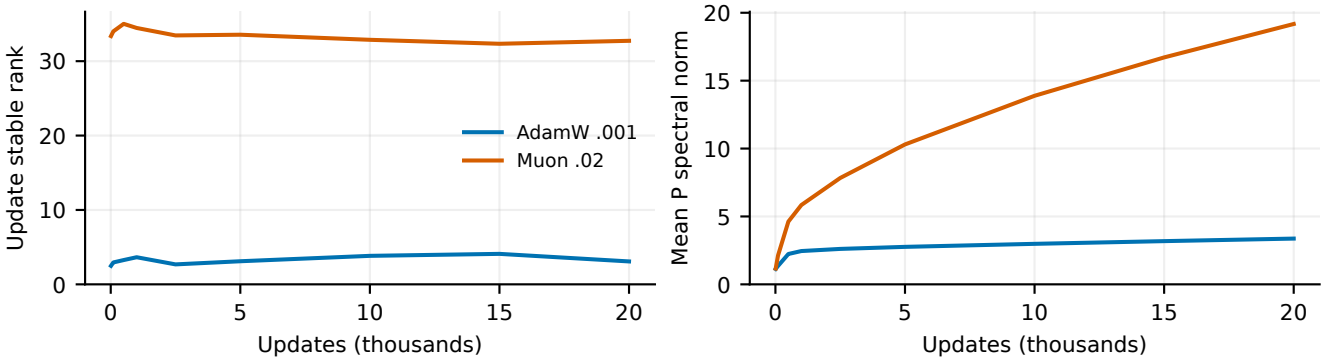


Figure 1: Product16 scalar K2, six-matrix means. Matrix rates: AdamW .001, Muon .02. Update stable rank is $\|\Delta P\|_F^2/\|\Delta P\|_2^2$. Muon broadens the update spectrum while P's spectral norm grows.

At 20k, AdamW/Muon update stable ranks are 3.08/32.74 and mean P norms 3.37/19.17. Field MSE improves from 0.03343 to 0.02385; SwiGLU192 reaches 0.01567. Norm-matched momentum does not reproduce all gains. Decay and training-time flattening harm fitting; flattening also worsens SW2.

3 From field structure to bounded features

Full-gradient readout. We tested whether structure in the complete predicted field matters more than a constraint on hidden P. For a vector backbone G, predict

$$v(x,t) = \nabla_x[\tfrac{1}{2}x^\top G(x,t)] = \tfrac{1}{2}[G(x,t) + J_G(x,t)^\top x]. \tag{2}$$

This matches the gradient structure of the analytic mixture velocities. At 5k, original-ODE Product16 MSE falls from 0.325527 to 0.037320; gradient SwiGLU reaches 0.033826 at nearly equal dense inference work. Local-adjoint controls lose much of the gain: the useful operation acts on the complete field and benefits both backbones.

Input-dependent directions. To change the directions in which P acts while preserving its pointwise spectrum, we tested $Q(h)T_K(s(h)P)Q(h)^\top h$ with an orthogonal learned frame. On a 2D mode plane embedded in 128D, frame K2 reaches MSE 0.000685 and K1 reaches 0.000713, versus SwiGLU192's 0.003095. Adding variance outside the plane reverses the sampling result: K2 gives 0.3173 versus SwiGLU192's 0.2471. Direction adaptation helps the thin geometry; extra Taylor order does not consistently help sampling.

Nonlinear internal flow. We tested whether linear dynamics were too restrictive by splitting $dy/dr = Py - y^{\odot 3}$ into linear steps and the exact cubic response $q_r(u) = u/\sqrt{1+2ru^2}$. At 5k on Product16, MSE is 0.092938 for tied cubic, 0.037159 for free ISRU and 0.078178 for SwiGLU160. Free ISRU uses independent matrices around q and a special initializer. Its advantage reverses on dependent-sign and half-mixture laws. Noisy-label, affine-head, frozen-feature and local-teacher probes leave part of the fitting gap unresolved.

Input-dependent features. This response motivates bounding the feature acted on by P. For $A(h) = PD(h)$ with $D(h) = \mathrm{diag}(d(h))$, the K1 map and Jacobian are

$$F_1(h) = h + P[d(h) \odot h], \qquad J_{F_1} = I + P[\mathrm{diag}(d) + \mathrm{diag}(h)J_d]. \tag{3}$$

We tested rank-matched linear/MLP controllers, sigmoid/affine outputs, direct diagonals and radial controls. Raw h helps Ring8 fitting; absolute and radial variants do not improve the overall tradeoff. ISRU-D uses $d_i = (1 + 1.2h_i^2)^{-1/2}$, giving $Pq_{0.6}(h)$ in each K1 increment. Its two maps surround SiLU; free ISRU uses $Aq_{0.6}(Bh)$. Their residual placement, nonlinear maps and initialization differ.

Ring8 FFN	Opt.	Field MSE	SW2, banks	SW2, pool
ISRU-D	A	**0.00977**	0.17110 (0.04721)	0.13261
Free ISRU	A	0.01000	0.17930 (0.04728)	0.14467
SwiGLU64	A	0.01372	0.19453 (0.06503)	0.17771
SwiGLU64	M	0.01128	0.17145 (0.05310)	0.13643
SwiGLU192	M	0.01123	0.18399 (0.05724)	0.15769

Table 1: 20k, Heun128. A: AdamW .0003; M: selected Muon .02. Banks: mean (SD) over eight common 4,096-sample banks. Pool: follow-up combining 32,768 saved samples. ISRU-D/free: 56,706 parameters, 55,616 dense MACs; Swi64: 56,766 and 55,232. MACs exclude pointwise work.

ISRU-D fits better than these four controls at 10k, 15k and 20k; sampling is close to selected Swi64. On Product16 at 20k/Heun128, eight-bank mean SW2 is 0.08900 for ISRU-D, 0.08596 for free ISRU and 0.08350 for selected SwiGLU192. The bounded response gives a task-dependent fitting benefit.

4 Checkpoint diagnosis: shape and mode allocation

Purpose and observation. To explain the fitting–sampling mismatch, we compared the same 15k/20k Ring8 checkpoints for ISRU-D, free ISRU and selected Swi64. ISRU-D MSE improves from 0.01026 to 0.00977 while eight-bank mean SW2 rises from 0.16054 to 0.17110. Pooling the 32,768 saved samples gives SW2 0.11727 to 0.13261. Assigning each sample to its nearest mode center shows two opposing changes: centered-residual SW2 improves from 0.01768 to 0.01502, but sector-frequency L1 error against equal weights grows from 0.03827 to 0.05450. All three models show better residual SW2 and worse sector balance.

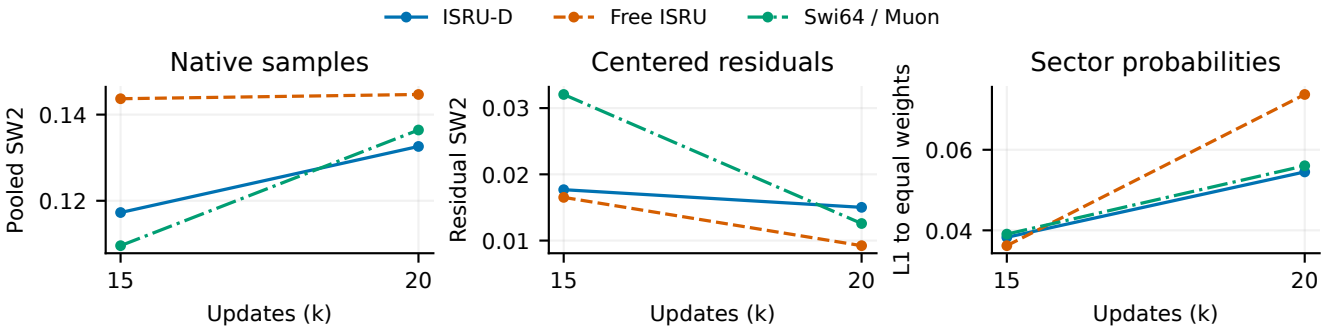


Figure 2: Ring8, Heun128, all eight banks pooled. Left: native SW2. Middle: SW2 after subtracting assigned mode centers. Right: sector-frequency L1 error against 1/8. Each panel has its own scale; the distances are not additive.

Where trajectories change. A common 4,096-source oracle path locates the discrepancy. Across the six checkpoints, 97.5–99.7% of observed sector crossings occur by $t = .75$. For ISRU-D at 20k, replacing the first 75% of the path with oracle dynamics reduces same-source endpoint MSE from 0.04934 to 0.00075. Keeping the learned prefix and replacing only the last quarter gives 0.05009. Most particle-to-mode differences are established early or midway. Median directional propagation gains remain close across checkpoints. The preselected trajectory bank improves its SW2 at 20k, opposite to the eight-bank mean; pooled and all leave-one-bank-out results accompany this bank-specific intervention.

A quantity to predict. Ring8 reflection symmetry makes the analytic velocity u tangent to every angular sector boundary. For learned density ρ_t transported by v, sector mass M_k obeys continuity:

$$\dot{M}_k(t) = -\int_{\partial S_k} \rho_t\, v \cdot n_k\, dS, \qquad B_k = -\int_0^1 \int_{\partial S_k} p_t(v-u)\cdot n_k\, dS\, dt. \tag{4}$$

Here p_t is the oracle density. For a small field perturbation $u + \epsilon(v-u)$, smooth density response gives $M_k^\epsilon(1) = 1/8 + \epsilon B_k + o(\epsilon)$. Net boundary flux, rather than gross crossings, changes mode balance. This signed boundary functional differs from bulk quadratic field error. For a K1 matrix change, $D_P v[\Delta P] = J_{\text{down}} \Delta P\, q(h)$: features and downstream derivatives connect P updates to the measured drift.

5 Hypotheses for the next two weeks

1. **Boundary flux predicts mode imbalance.** Compute B_k for the six frozen fields and calibrate it under weak oracle/model interpolation. Agreement with measured sector drift selects a boundary-sensitive dynamics control; failure selects nonlinear density response for further diagnosis.
2. **Functional updates explain optimizer behavior.** Compare norm-matched P updates through $J_{\text{down}} \Delta P q$, field loss and predicted mass change. Match initialization and residual placement against free ISRU. Retain an ODE constraint only if it improves the measured tradeoff against that control; otherwise develop the shared feature response.